Answer the questions in the spaces provided on the question sheets. If you run out of room for an answer, continue on the back of the page.

Name: _____

Student ID: _____

Quiz 6: 09/24 Content

MATH 1271

09/29/2026

This quiz is worth 5 points and consists of 1 questions. You have 5 minutes to complete the quiz.

Question	Points
1	5
Total:	5

1. (5 points) Use the definition of the derivative to find the derivative of the function:

$$R(w) = w^2 - 8w + 20$$

Solution: We compute:

$$\begin{aligned}
R'(w) &= \lim_{h\to 0} \frac{R(w+h) - R(w)}{h} && (1)\\
&= \lim_{h\to 0} \frac{(w+h)^2 - 8(w+h) + 20 - (w^2 - 8w + 20)}{h} && (2)\\
&= \lim_{h\to 0} \frac{w^2 + 2wh + h^2 - 8w - 8h + 20 - w^2 + 8w - 20}{h} && (3)\\
&= \lim_{h\to 0} \frac{2wh + h^2 - 8w}{h} && (4)\\
&= \lim_{h\to 0} (2w + h - 8) && (5)\\
&= 2w + 0 - 8 = 2w - 8 && (6)
\end{aligned}$$

So $R'(w) = 2w - 8$.